

Basic OOP

1. What are the four pillars of OOP? Explain with examples.
2. What is the difference between abstraction and encapsulation?
3. Can we achieve multiple inheritance in Java? How?
4. What is method overloading vs method overriding?
5. What happens if we override a final method?
6. What is the difference between abstract class and interface?
7. What is dynamic method dispatch?
8. What is the difference between `==` and `.equals()`?
9. What is the difference between shallow copy and deep copy?
10. What are immutable classes? How do you create one?
11. What is the use of super keyword?
12. What is covariant return type?
13. Explain composition vs inheritance.
14. What is the difference between `this()` and `super()`?
15. What are marker interfaces? Give examples.

Functional Programming

1. What is a functional interface?
2. What are lambda expressions? Why were they introduced?
3. What is the difference between lambda and anonymous class?
4. What is the invokedynamic instruction?
5. What is a method reference? Types?
6. What is the Stream API?
7. Difference between `map()` and `flatMap()`?
8. What is the difference between intermediate and terminal operations?
9. What is Optional? Why is it used.
10. What are built-in functional interfaces?
11. What is default method in interface?
12. What happens internally when you write a lambda?
13. What happens if two interfaces have same default method?
14. Can a functional interface have static methods?
15. What is the difference between Thread class and Runnable interface?